INCH-POUND

MIL-DTL-28748/20
w/AMENDMENT 1
7 July 2014
SUPERSEDING
MIL-DTL-28748/20
11 March 2005

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RACK AND PANEL, PRINTED WIRING BOARD (PWB) MOUNT, RIGHT ANGLE, PIN CONTACTS, SIZE 22

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28748.

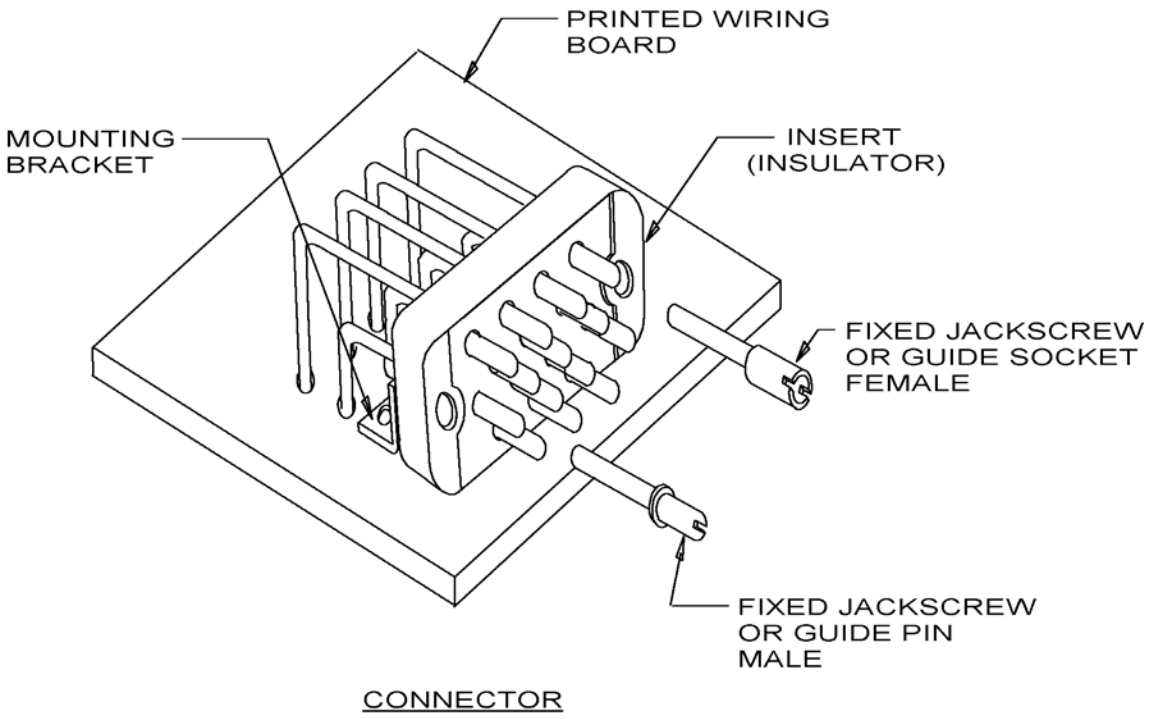
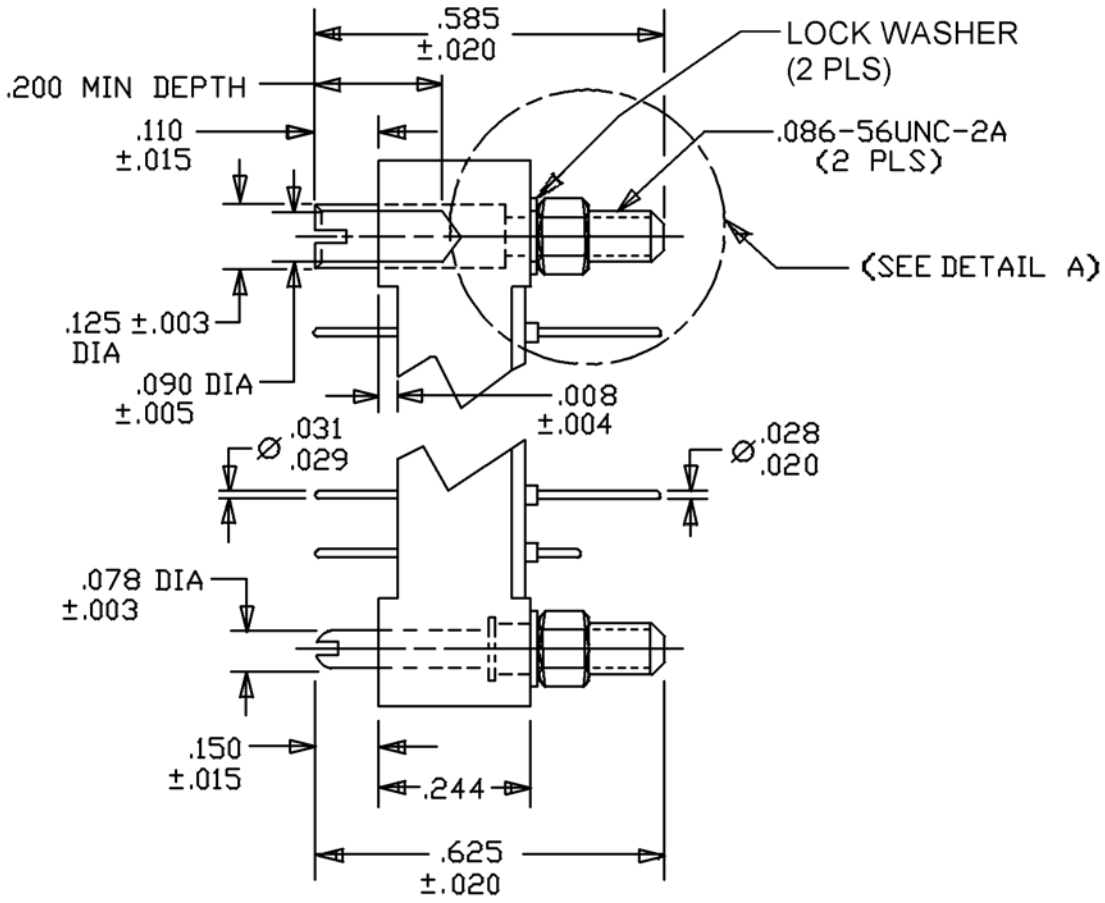


FIGURE 1. Connector with mounting bracket and jackscrews or guidepins.

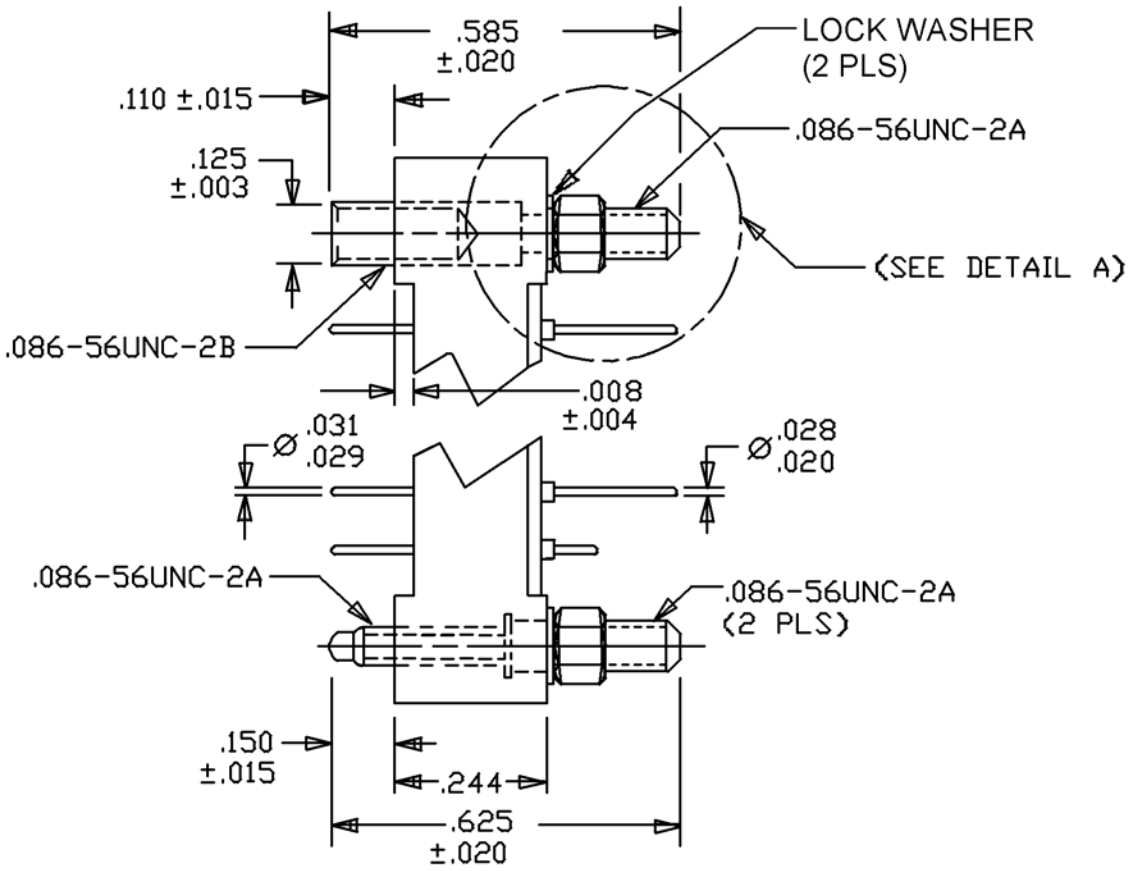
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CONNECTOR WITH GUIDEPINS
CONFIGURATION A

FIGURE 1. Connector with jackscrews and guidepins - Continued.

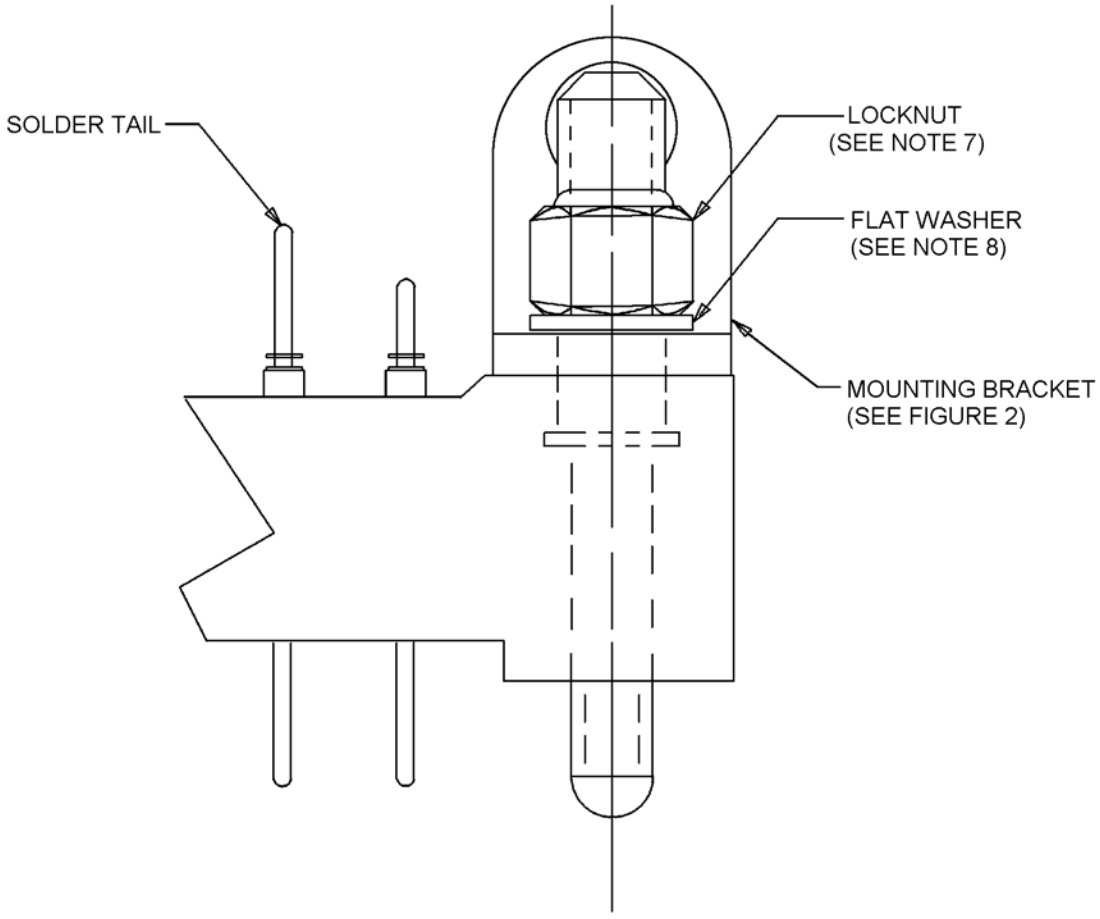
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CONNECTOR WITH JACKSCREWS
CONFIGURATION B

FIGURE 1. Connector with jackscrews and guidepins - Continued.

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DETAIL A

FIGURE 1. Connector with jackscrews and guidepins - Continued.

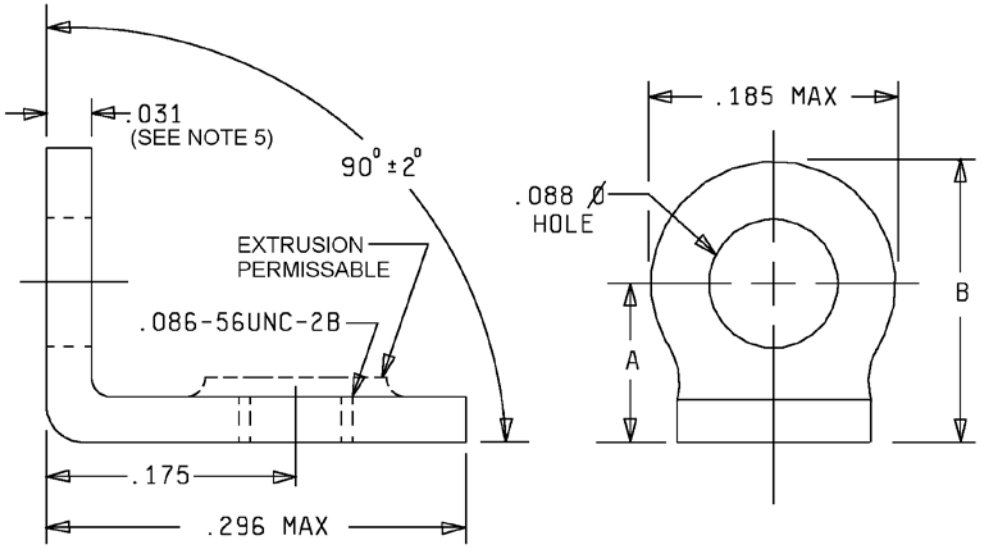
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Inches	mm	Inches	mm
.003	0.08	.078	1.98
.004	0.10	.086	2.18
.005	0.13	.090	2.29
.008	0.20	.110	2.79
.015	0.38	.125	3.18
.020	0.51	.150	3.81
.028	0.71	.200	5.08
.029	0.73	.244	6.20
.031	0.79	.585	14.86
		.625	15.88

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances is ± .020 inch (0.50 mm).
4. Materials for guidepins and jackscrews shall be corrosion resistant steel in accordance with ASTM A276 class 304, 316, or ASTM A582/A582M, class 302, 303, 303Se.
5. Washer .020 inch (0.51 mm) may be aluminum or corrosion resistant steel.
6. Finish for washers shall be either passivated or anodized.
7. Locknut: Mounting hardware shall include stainless steel locknuts. The nuts may be all metal or designed with nylon inserts.
8. Flat washer .020 inch (0.51 mm) thick supplied with all sizes.

FIGURE 1. Connector with jackscrews and guidepins - Continued.

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MOUNTING BRACKET

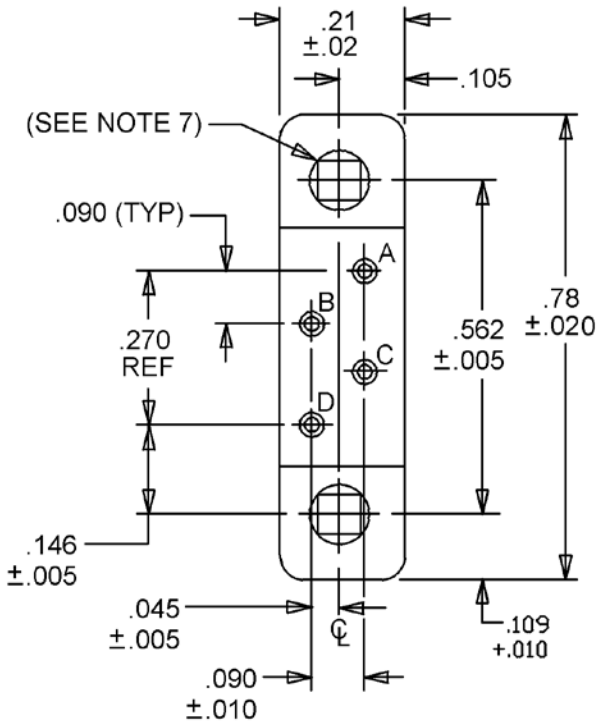
Inches	mm
.031	0.79
.086	2.18
.088	1.24
.175	4.45
.185	4.70
.296	7.52

Part identifying number (PIN) designator	Dimension A	Dimension B (Max)	Insert size
1 (see note 5)	.105 ±.007 (2.67 ± 0.18))	.205 (5.21)	4 and 7
2	.140 (3.56)	.240 (6.10)	14, 20, and 26
3	.195 (4.95)	.295 (7.49)	34, 44, and 50

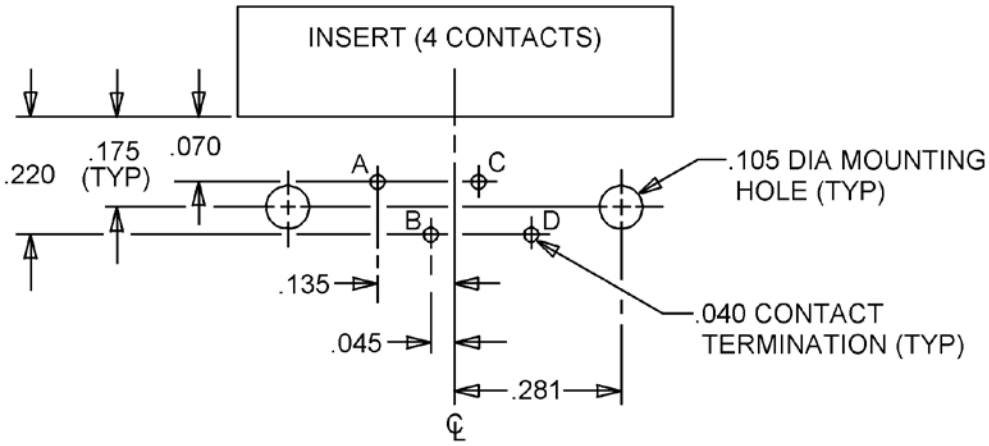
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerance is ± .005 inch (0.13 mm).
4. Material: Aluminum alloy in accordance with SAE-AMS-QQ-A-250/8.
5. Dash 1 only. Material thickness dimension may be .024 inch (0.61 mm) minimum.
6. Bracket design optional, design shall fall within drawing parameters.

FIGURE 2. Mounting bracket.

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INSERT ARRANGEMENT (4 CONTACTS)

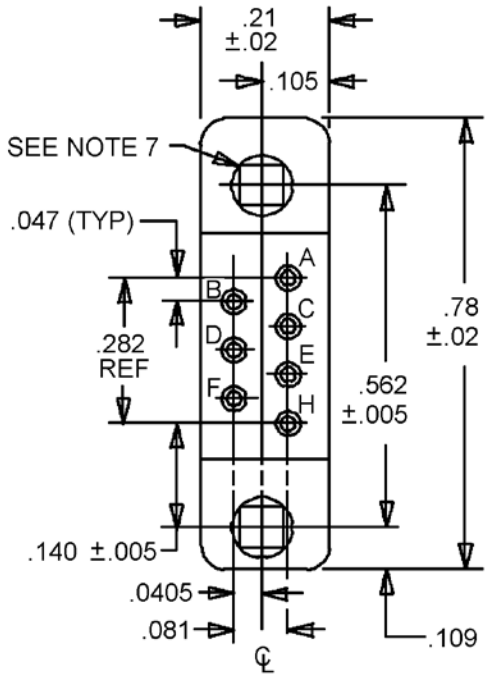


RECOMMENDED MOUNTING PATTERN (4 CONTACTS)

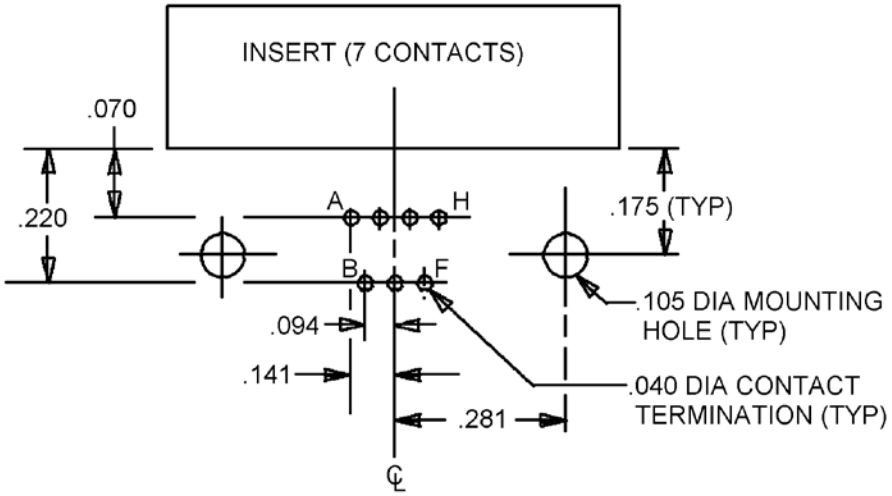
CONFIGURATION A

FIGURE 3. Insert arrangements.

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INSERT ARRANGEMENT (7 CONTACTS)

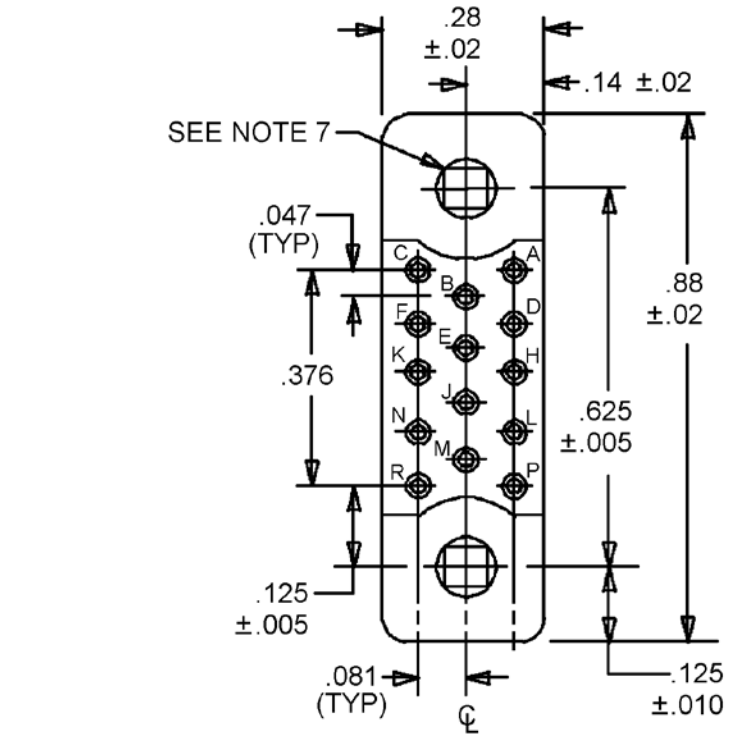


RECOMMENDED MOUNTING PATTERN (7 CONTACTS)

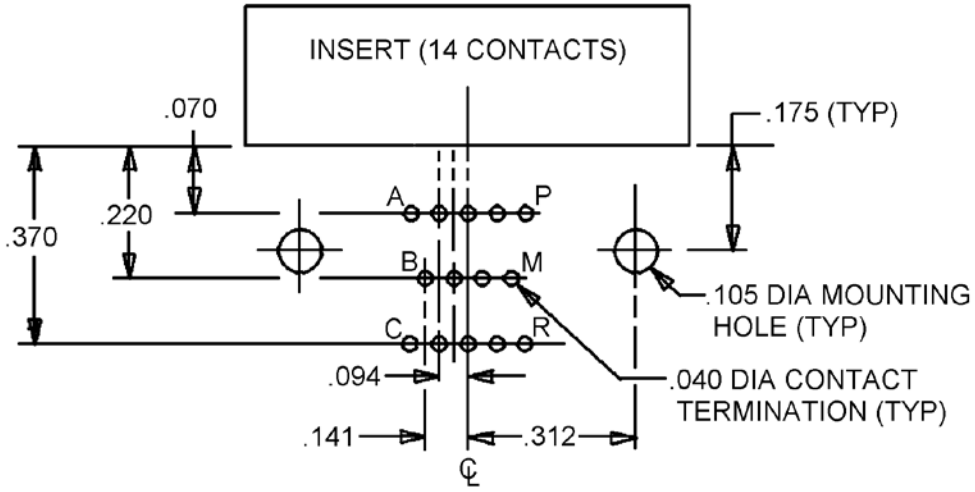
CONFIGURATION B

FIGURE 3. Insert arrangements - Continued.

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INSERT ARRANGEMENT (14 CONTACTS)

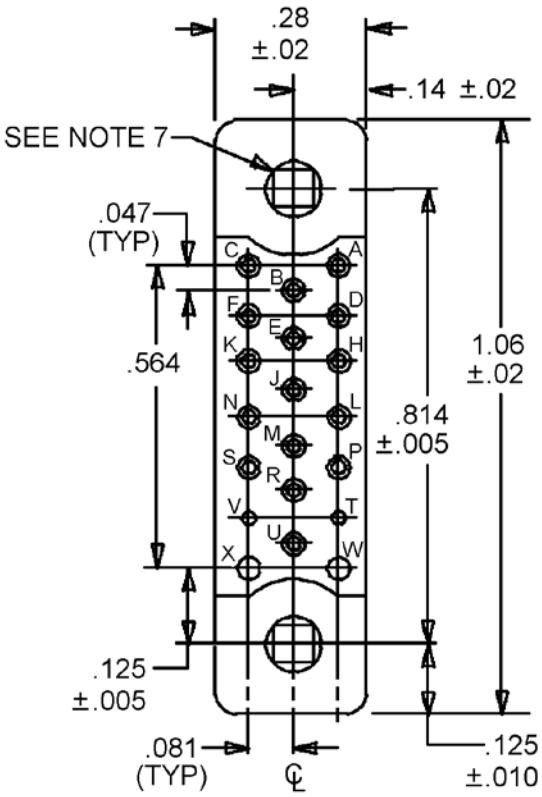


RECOMMENDED MOUNTING PATTERN (14 CONTACTS)

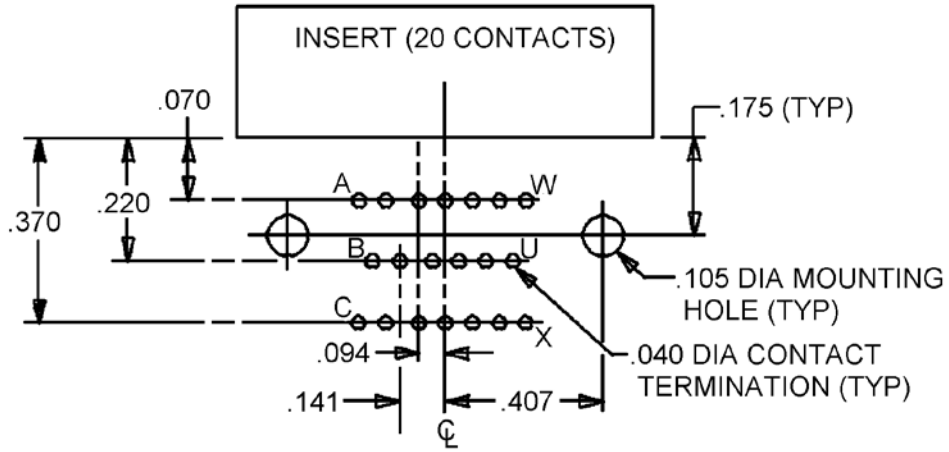
CONFIGURATION C

FIGURE 3. Insert arrangements - Continued.

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INSERT ARRANGEMENT (20 CONTACTS)



RECOMMENDED MOUNTING PATTERN (20 CONTACTS)

CONFIGURATION D

FIGURE 3. Insert arrangements - Continued.

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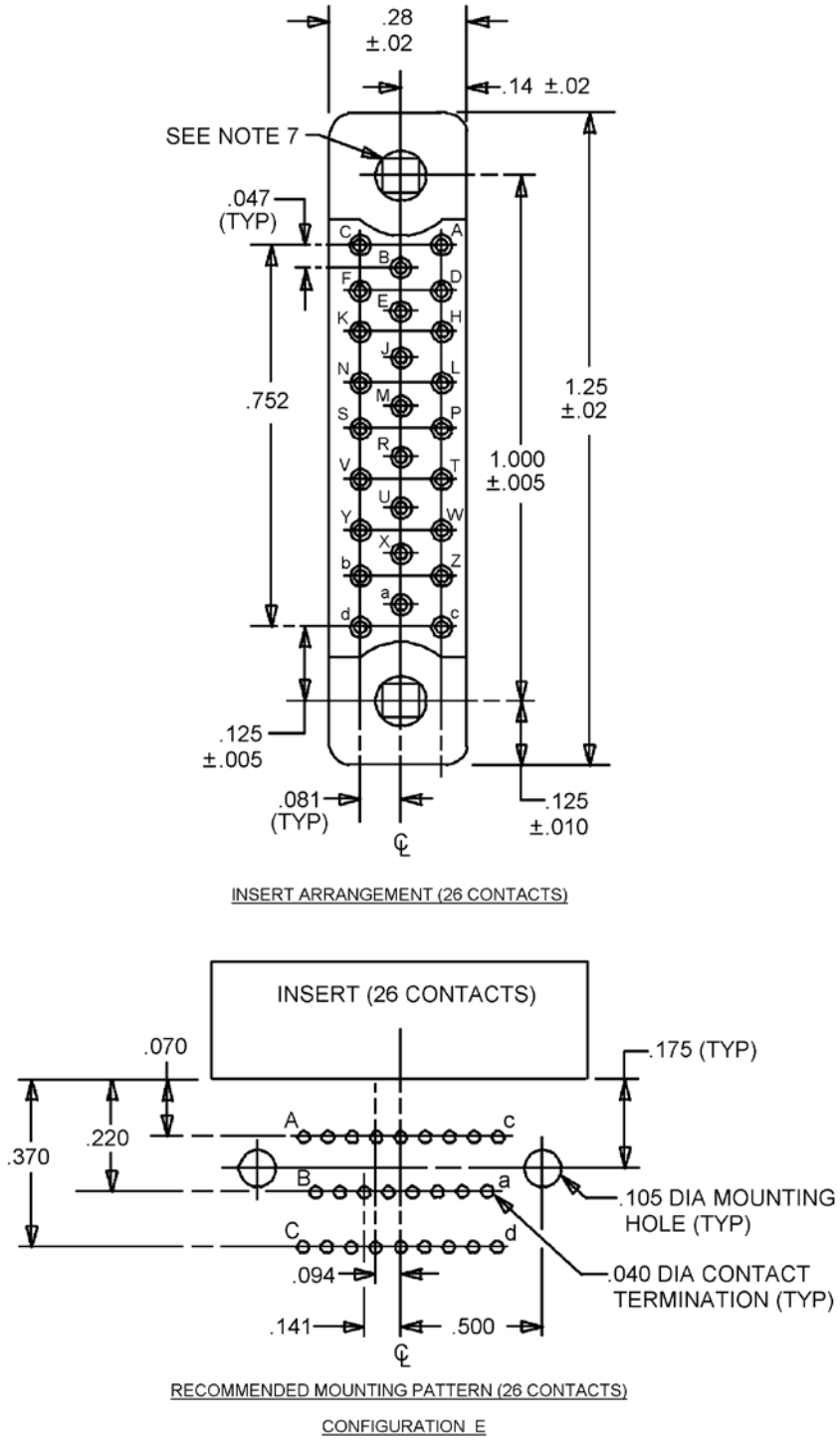
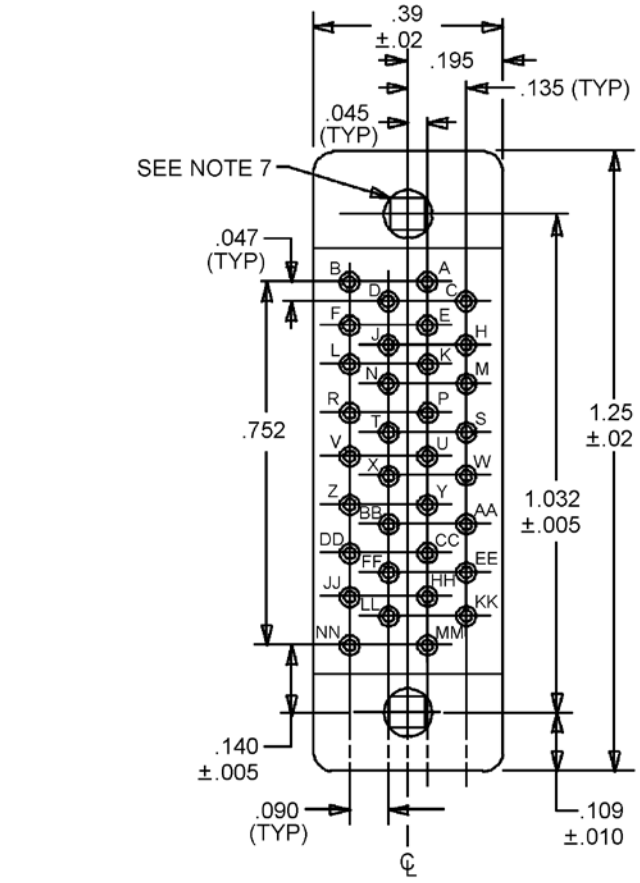


FIGURE 3. Insert arrangements - Continued.

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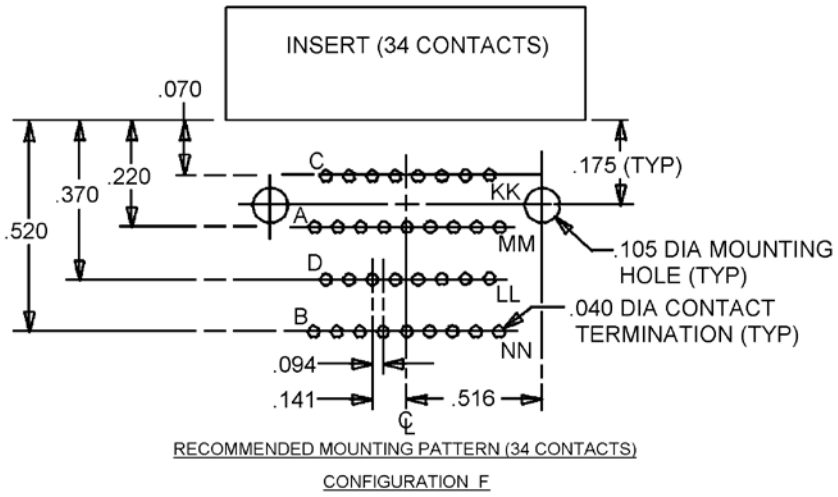
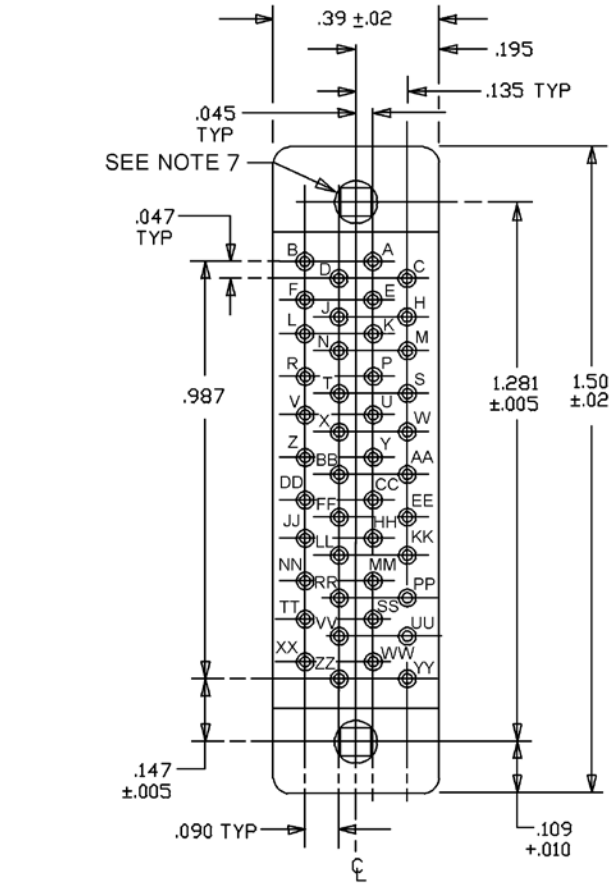


FIGURE 3. Insert arrangements - Continued.

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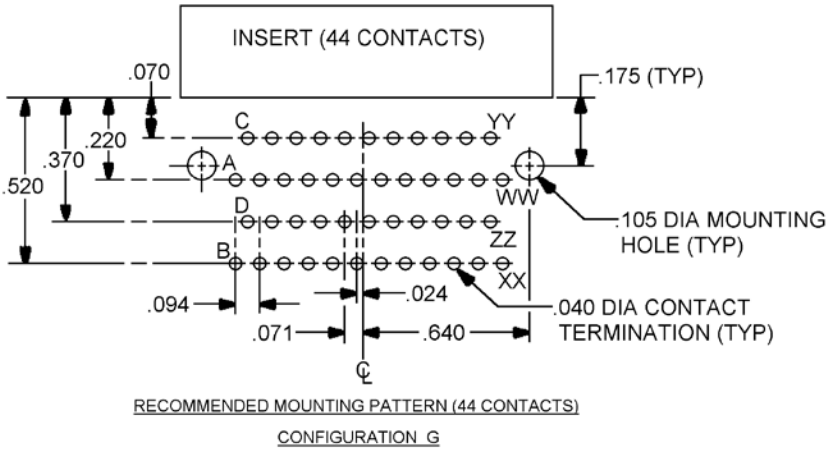


FIGURE 3. Insert arrangements - Continued.

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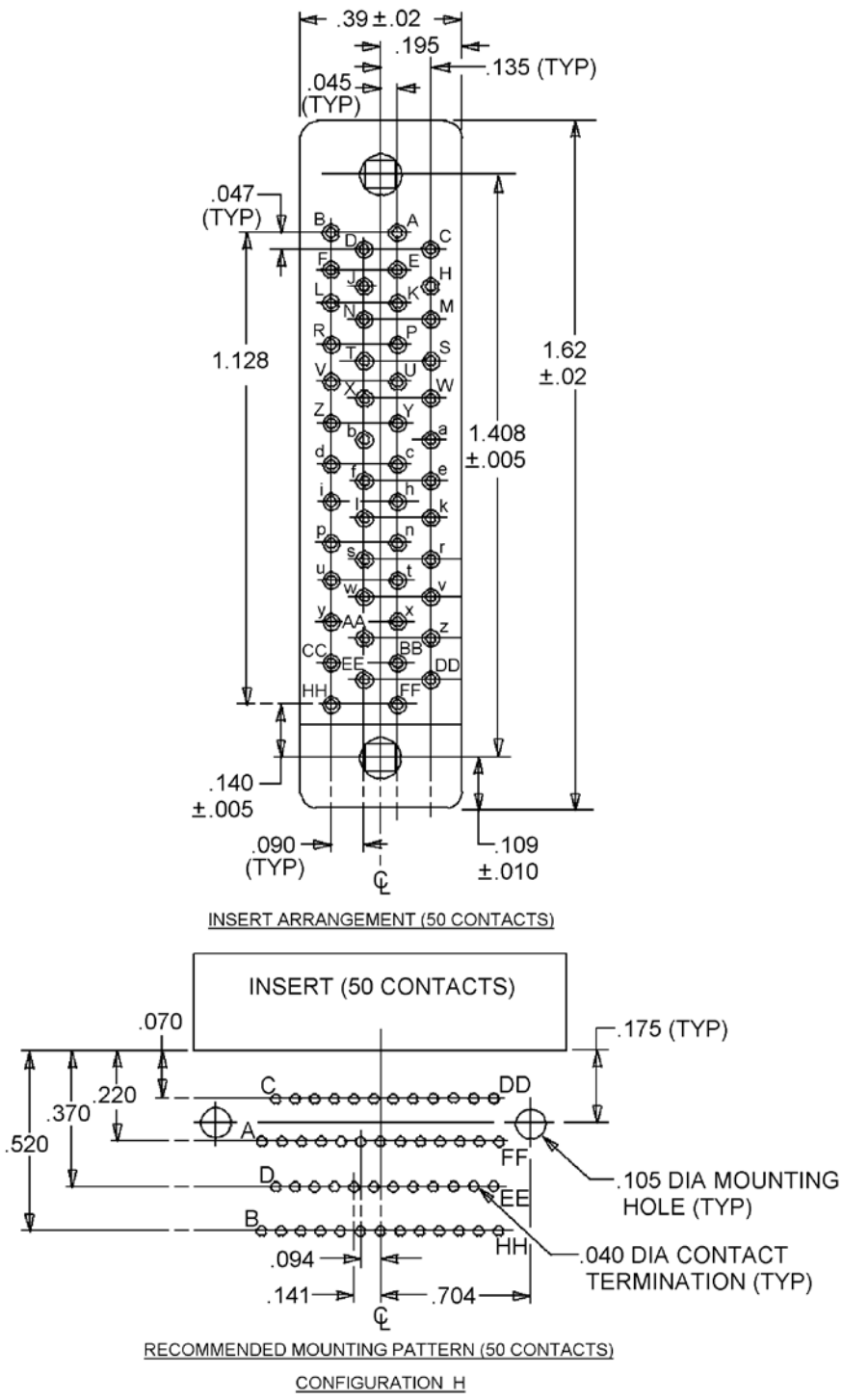
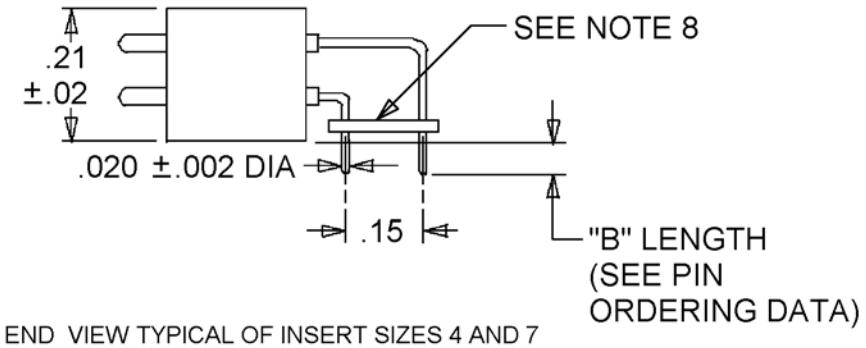
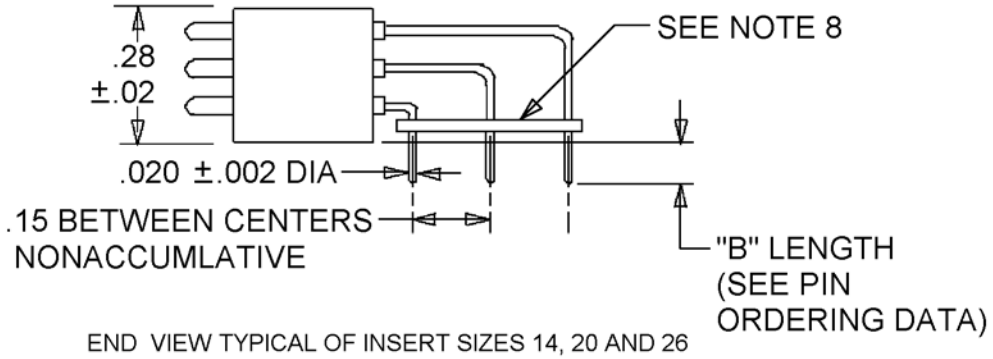


FIGURE 3. Insert arrangements - Continued.

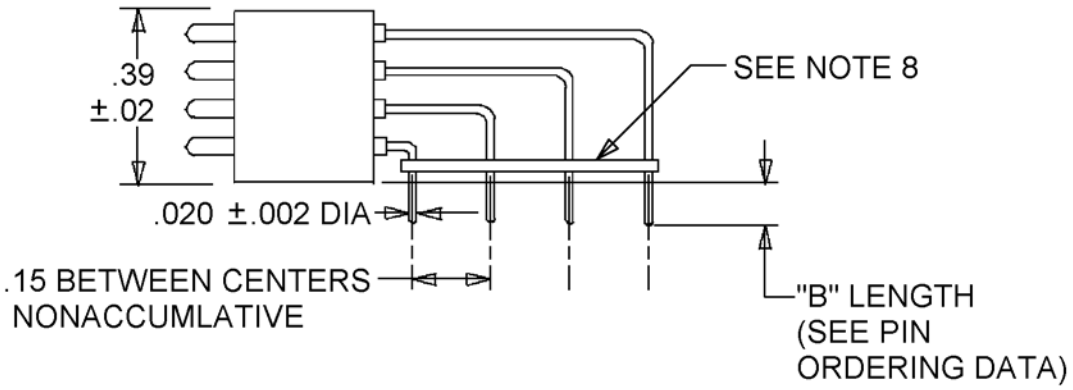
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END VIEW TYPICAL OF INSERT SIZES 4 AND 7



END VIEW TYPICAL OF INSERT SIZES 14, 20 AND 26



END VIEW TYPICAL OF INSERT SIZES 34, 44 AND 50

FIGURE 3. Insert arrangements - Continued.

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Inches	mm	Inches	mm
.002	0.05	.28	7.1
.005	0.13	.281	7.13
.010	0.25	.282	7.16
.02	0.5	.312	7.92
.020	0.51	.370	9.40
.024	0.61	.376	9.55
.040	1.02	.39	9.91
.0405	1.029	.407	10.33
.045	1.14	.500	12.70
.047	1.19	.516	13.11
.070	1.78	.520	13.21
.071	1.80	.562	14.27
.081	2.56	.564	14.33
.090	2.29	.625	15.87
.094	2.39	.640	16.26
.105	2.67	.704	17.88
.109	2.76	.752	19.10
.125	3.18	.78	19.8
.135	3.43	.814	20.68
.14	3.5	.88	22.4
.140	3.56	.987	25.06
.141	3.58	1.000	25.40
.146	3.71	1.032	26.21
.147	3.73	1.408	35.76
.15	3.81	1.06	26.9
.175	4.44	1.25	31.8
.195	4.95	1.128	28.65
.21	5.3	1.281	32.54
.220	5.59	1.50	38.1
.270	6.86	1.62	41.15

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerance is ±.015 inch (0.38 mm).
4. Tolerance between any two adjacent contact centers shall be ±.004 inch (0.10 mm). Tolerance between any two contact centers, other than adjacent centers, shall be ±.006 inch (0.15 mm).
5. Solder tail nonfunctional areas need not be plated, provided they do not cause degradation in performance.
7. Hardware hole configuration optional, design shall incorporate an anti-rotation feature on either front or rear face of insulator.
8. Manufacturer may include an optional .035 ±.010 inch (0.89 ±0.25 mm) plastic alignment bar of nylon or polyphenylene sulfide in order to ensure the proper alignment or terminals.

FIGURE 3. Insert arrangements - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figures 1, 2, and 3.

Mating connector: See MIL-DTL-28748/8.

Contacts: Contacts shall be in accordance with MIL-DTL-28748.

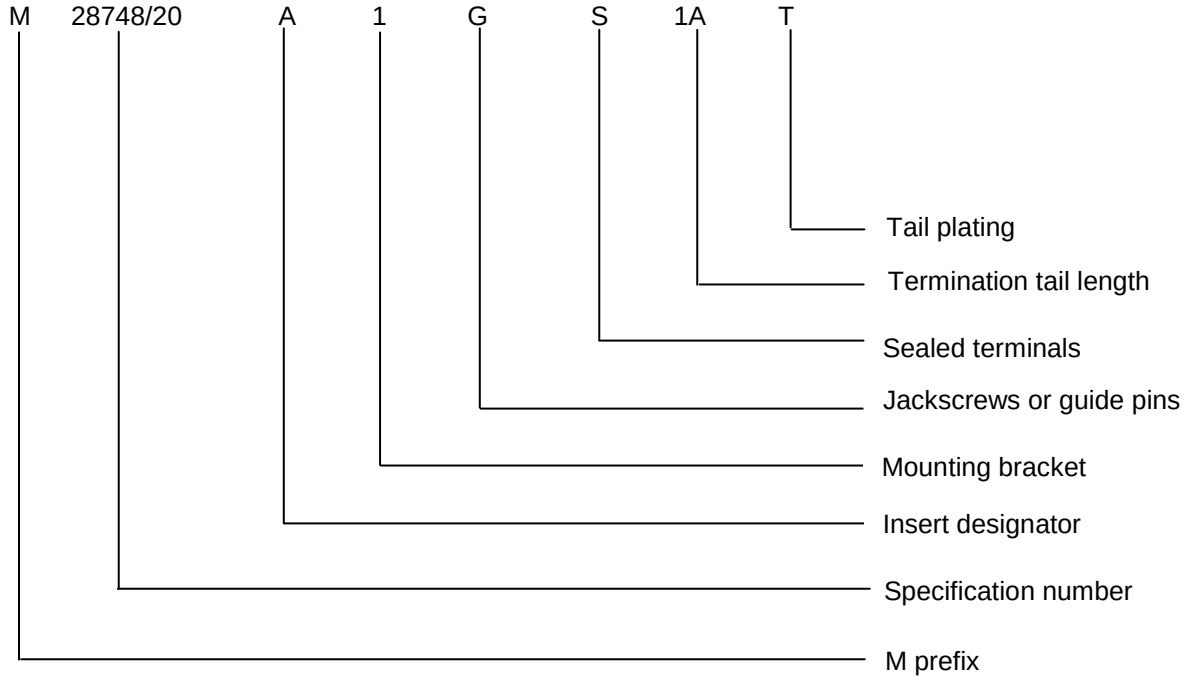
Solder tail: "B" length (see permissible PINS).

Solder tail finish:

a. Tin/lead in accordance with SAE-AMS-P-81728, 50 to 97 percent tin, 100 micro inches (2.54µm) minimum or hot solder dip in accordance with J-STD -006 63 percent tin, 100 micro inches (2.54µm) minimum over 50 to 100 microinches (1.27 to 2.54µm) nickel in accordance with SAE-AMS-QQ-N-290, class 1.
b. Gold in accordance with MIL-DTL-45204, type II, grade C, class 0 or equivalent over 50 to 100 microinches (1.27 to 2.54µm) of nickel in accordance with SAE-AMS-QQ-N-290, class 1.

Contact sealants: Sealants used to back seal contact cavities to prevent entry of cleaning solutions, conformal coatings, etc.

Part or Identifying Number (PIN): As shown in the following example: The number zero (0) is used to indicate which parts are not included (see table I permissible PINS).



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Insert designator		Number of contact positions
A	-	4
B	-	7
C	-	14
D	-	20
E	-	26
F	-	34
G	-	44
H	-	50

Mounting bracket (see figure 2): To order separately see table I permissible PINs.

Mounting bracket Designator		Insert size
1	-	4 and 7
2	-	14, 20, and 26
3	-	34, 44, and 50
-0	-	None

Jackscrews or guidepins:

F - fixed jackscrews, configuration B, figure 1.
G - guidepins, configuration A figure 1.
0 - none included.

Jack and guide location: The jackscrew female (turning or fixed) or guide socket shall be located by the "A" socket contact.

Seal: Manufacturer may supply sealed terminals when no seal is specified.

S - sealed
0 – No seal

Termination tails "B" length (see figure 3):

1A - Right angle - solder tails .093 (2.76 mm) inch.
1B - Right angle - solder tails .125 (3.18 mm) inch.
1C - Right angle - solder tails .156 (3.96 mm) inch.

Tail plating:

T - Tin/lead
G - Gold

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Permissible PINs are specified in table I.

TABLE I. Permissible PINs. 1/

M prefix and specification number	Insert designator	Mounting bracket designator	Jackscrews or guide pins	Sealed terminals	Solder tail plating	Solder tail length
M28748/20	A	1 or 0	F - fixed G - Guide pin 0 - none	S - sealed 0 - no seal	T - tin/lead G - gold	1A 1B 1C
	B	1 or 0				
	C	2 or 0				
	D	2 or 0				
	E	2 or 0				
	F	3 or 0				
	G	3 or 0				
	H	3 or 0				

Mounting bracket only 2/	Insert designator	Mounting bracket designator
M28748/20	-0	1, 2, 3, 4

1/ The number zero (0) is used to indicate which parts are not included.
2/ Mounting brackets are supplied in pairs.

Referenced documents: In addition to MIL-DTL-28748, this document references the following:

MIL-DTL-28748/8
MIL-DTL-45204
ASTM A582/A582M
ASTM A276
J-STD-006
SAE-AMS-P-81728
SAE-AMS-QQ-A-250/8
SAE-AMS-QQ-N-290

CONCLUDING MATERIAL

Custodians:
Army -CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Air Force - 99

Preparing activity:
DLA - CC

(Project 5935-2014-029)

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